Sensitive Periods for L2 Learning

L2 learning as a model system for sensitive period research

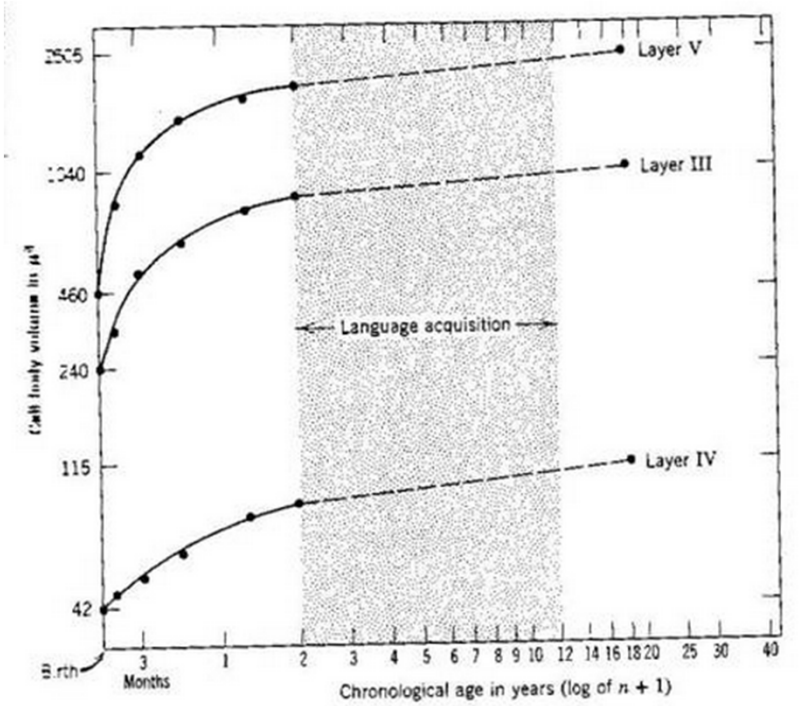


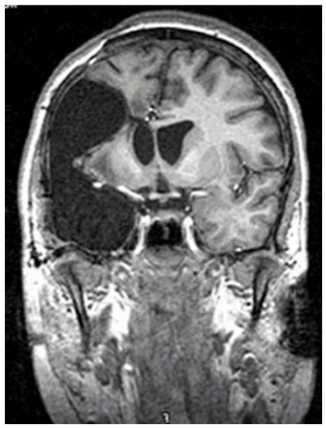
FIG. 4.5. Neurons in the cerebral cortex grow in volume with age. The broken lines between age two and eighteen indicate that no measurements were made on brains of children who died between these ages. (Redrawn from Schadé and van Groenigen, 1961.)

Remember that Lenneberg (1967) developed an influential theory that the sensitive period had to do with the development of cerebral dominance, or developmental changes in language laterality.

This was motivated by

- a rough alignment of the literature on brain development with new insights about language milestones
- a review of the literature on resilience after brain injury or surgery

L2 learning as a model system for sensitive period research





Much of what is known about L1 sensitive periods is from research on idiosyncratic cases, or relatively rare phenomena.

In contrast, many people learn second languages at different times in their lives, often with somewhat limited success. Those limits have also been used to reason about the timing of the sensitive/critical period.

Critical Period Effects in Second Language Learning: The Influence of Maturational State on the Acquisition of English as a Second Language

JACQUELINE S. JOHNSON AND ELISSA L. NEWPORT

University of Illinois

Lenneberg (1967) hypothesized that language could be acquired only within a critical period, extending from early infancy until puberty. In its basic form, the critical period hypothesis need only have consequences for first language acquisition. Nevertheless, it is essential to our understanding of the nature of the hypothesized critical period to determine whether or not it extends as well to second language acquisition. If so, it should be the case that young children are better second language learners than adults and should consequently reach higher levels of final proficiency in the second language. This prediction was tested by comparing the English proficiency attained by 46 native Korean or Chinese speakers who had arrived in the United States between the ages of 3 and 39, and who had lived in the United States between 3 and 26 years by the time of testing. These subjects were tested on a wide variety of structures of English grammar, using a grammaticality judgment task. Both correlational and *t*-test analyses demonstrated a clear and strong advantage for earlier arrivals over the later arrivals. Test performance was linearly related to age of arrival up to puberty; after puberty, performance was low but highly variable and unrelated to age of arrival. This age effect was shown not to be an inadvertent result of differences in amount of experience with English, motivation, self-consciousness, or American identification. The effect also appeared on every grammatical structure tested, although the structures varied markedly in the degree to which they were well mastered by later learners. The results support the conclusion that a critical period for language acquisition extends its effects to second language acquisition. © 1989 Academic Press, Inc.

(1a) The farmer bought two pigs at the market.
*(1b) The farmer bought two pig at the market.
(2a) The little boy is speaking to a policeman.
*(2b) The little boy is speak to a policeman.

(6a) Tom is reading a book in the bathtub.
*(6b) Tom is reading book in the bathtub.
(7a) The boys are going to the zoo this Saturday.
*(7b) A boys are going to the zoo this Saturday.
(8a) Larry went home after the party.
*(8b) Larry went the home after the party.

(23a) When will Sam fix his car?
*(23b) When Sam will fix his car?
(24a) What do they sell at the corner store?
*(24b) What they sell at the corner store?

Johnson & Newport devised a grammaticality judgment test (GJT)

Linguistics quiz: Can you determine what is being tested in each of these example sets?

Participants listened to recorded sentences and simply had to say whether each sentence was grammatical or not.

The test was untimed.

The data

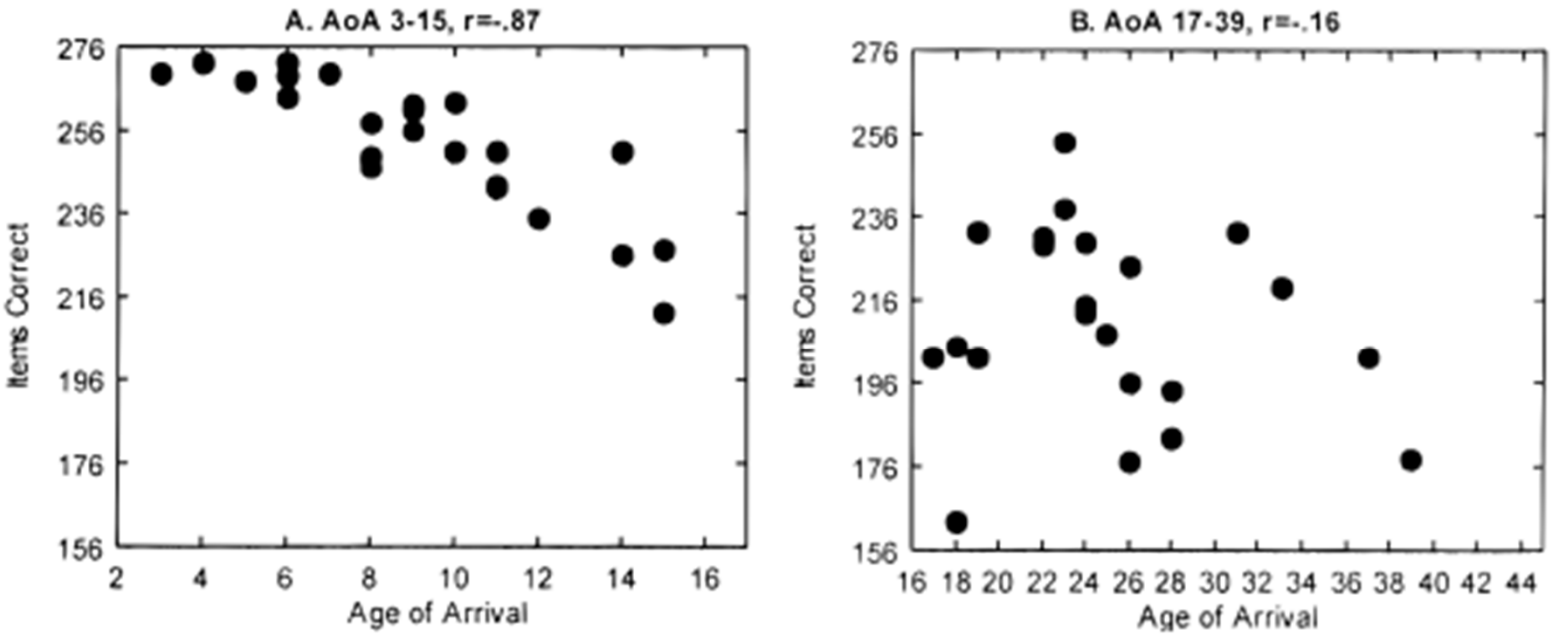


FIG. 1. The age function in J&N89, Early Arrivals (A), and Late Arrivals (B). In this and Figs. 2 and 3, points represent number of items correct for individual participants.

What patterns do we see in these data?

The data

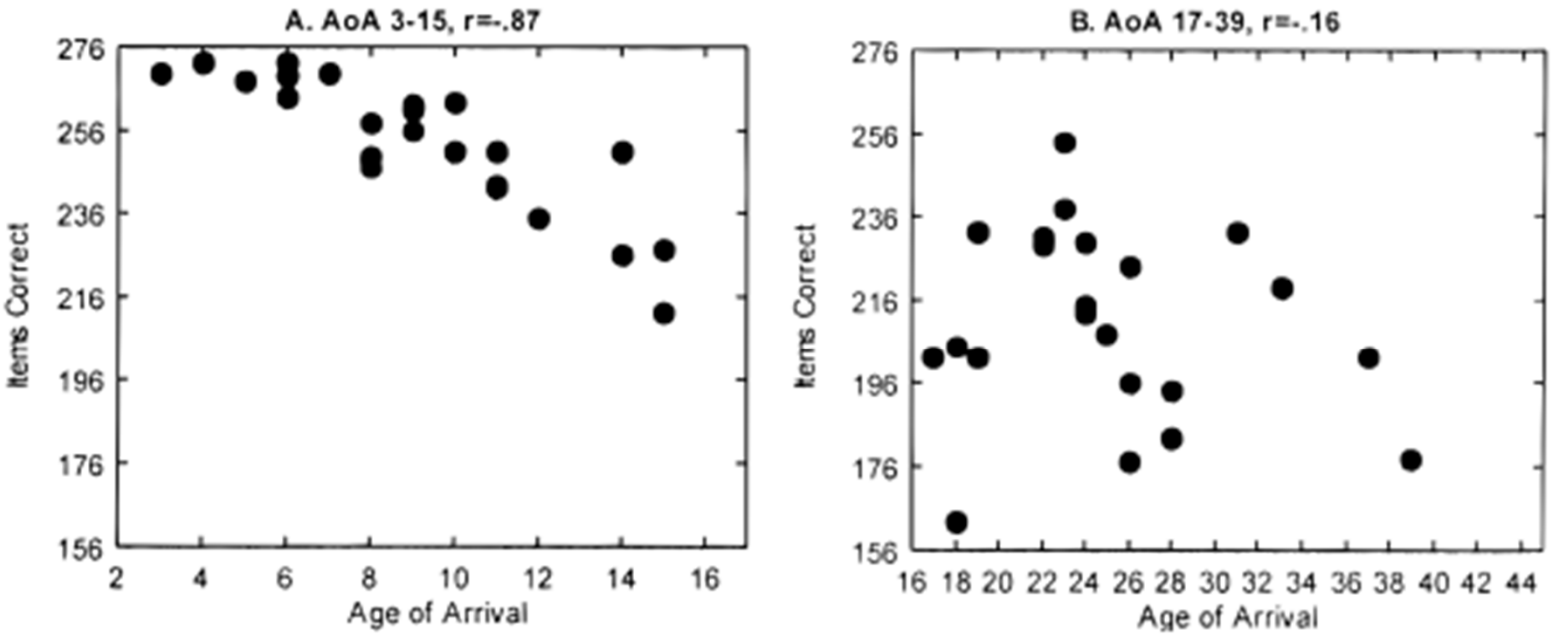


FIG. 1. The age function in J&N89, Early Arrivals (A), and Late Arrivals (B). In this and Figs. 2 and 3, points represent number of items correct for individual participants.

- strong relationship between AOA and accuracy
- early learners (up to about 8) are often near-perfect (native-like)
- nearly random distribution of accuracy for Late Arrivals

Johnson & Newport argue that this is consistent with a sensitive period (in fact, they say "critical period") for L2 learning.

First, the dependence of ulitmate attainment on AOA is consistent with a loss of language learning ability over development.

(The timing is not entirely consistent with Lenneberg's hypothesis, but remember that he was working with a large gap between early childhood and adolescence.)

Second, attainment in late acquisition is all over the place.

After the closure of the sensitive period, they argue, attainment depends on idiosyncratic learning abilities, not the usual language acquisition processes.

Two versions of the “critical period” hypothesis

Version One: The exercise hypothesis. Early in life, humans have a superior capacity for acquiring languages. If the capacity is not exercised during this time, it will disappear or decline with maturation. If the capacity is exercised, however, further language learning abilities will remain intact throughout life.

Version Two: The maturational state hypothesis. Early in life, humans have a superior capacity for acquiring languages. This capacity disappears or declines with maturation.

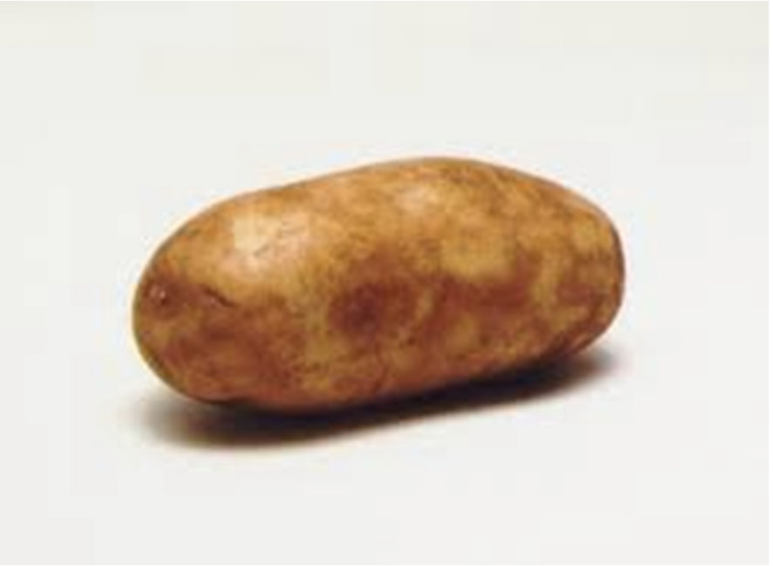
Which of these do you think JN89 see as more consistent with their data?

Does this change our interpretation of the L1 critical period data?

Methodological issue: Test design

The Grammatical Judgment Task is very different from fluent language use.

One thing people worry about is the use of “metalinguistic” or “explicit” knowledge.



This is to remind me to tell you a story about “potato” and allophones.

Table 1. Key characteristics of implicit and explicit knowledge

Characteristics	Implicit knowledge	Explicit knowledge
Awareness	Intuitive awareness of linguistic norms	Conscious awareness of linguistic norms
Type of knowledge	Procedural knowledge of rules and fragments	Declarative knowledge of grammatical rules and fragments
Systematicity	Variable but systematic knowledge	Anomalous and inconsistent knowledge
Accessibility	Access to knowledge by means of automatic processing	Access to knowledge by means of controlled processing
Use of L2 knowledge	Access to knowledge during fluent performance	Access to knowledge during planning difficulty
Self-report	Nonverbalizable	Verbalizable
Learnability	Potentially only within critical period	Any age

(Ellis, 2005, SSLA)

Methodological issue: Test design

Table 4. Design features of the tests

Criterion	Imitation	Oral narrative	Timed GJT	Untimed GJT	Metalanguage
Degree of awareness	Feel	Feel	Feel	Rule	Rule
Time available	Pressured	Pressured	Pressured	Unpressured	Unpressured
Focus of attention	Meaning	Meaning	Form	Form	Form
Metalinguistic knowledge	No	No	No	Yes	Yes

One suggestion is that the task demands of the untimed GJT make it a better measure of explicit learning of "metalinguistic" knowledge than grammar acquisition.

Consistent with this, untimed GJT performance is strongly correlated with years of formal education in L2.

Table 14. Correlations between starting age and years of formal instruction and test measures

	Variables	
Test	Starting age[a]	Formal instruction[a]
Imitation	−.11	.10
Oral narrative	.03	−.05
Timed GJT	−.23*	.17
Untimed GJT (grammatical items)	−.05	.03
Untimed GJT (ungrammatical items)	−.02	.27*
Metalinguistc knowledge	.05	.18

[a] In years.
*$p < .05$

(Ellis, 2005, SSLA)

Methodological/theoretical issue: transfer



auto



rivière



bicyclette



コロッケ (korokke)



モノレール (monoreru)

Remember we talked a lot about *transfer* in L2 learning.

Languages that are typologically more similar can benefit more from transfer.

Johnson & Newport tested people with English as L2 and Chinese or Korean as L1.

What do you suppose would happen if the languages were more similar?

(Ellis, 2005, SSLA)

Not all aspects of the grammar are equally difficult.

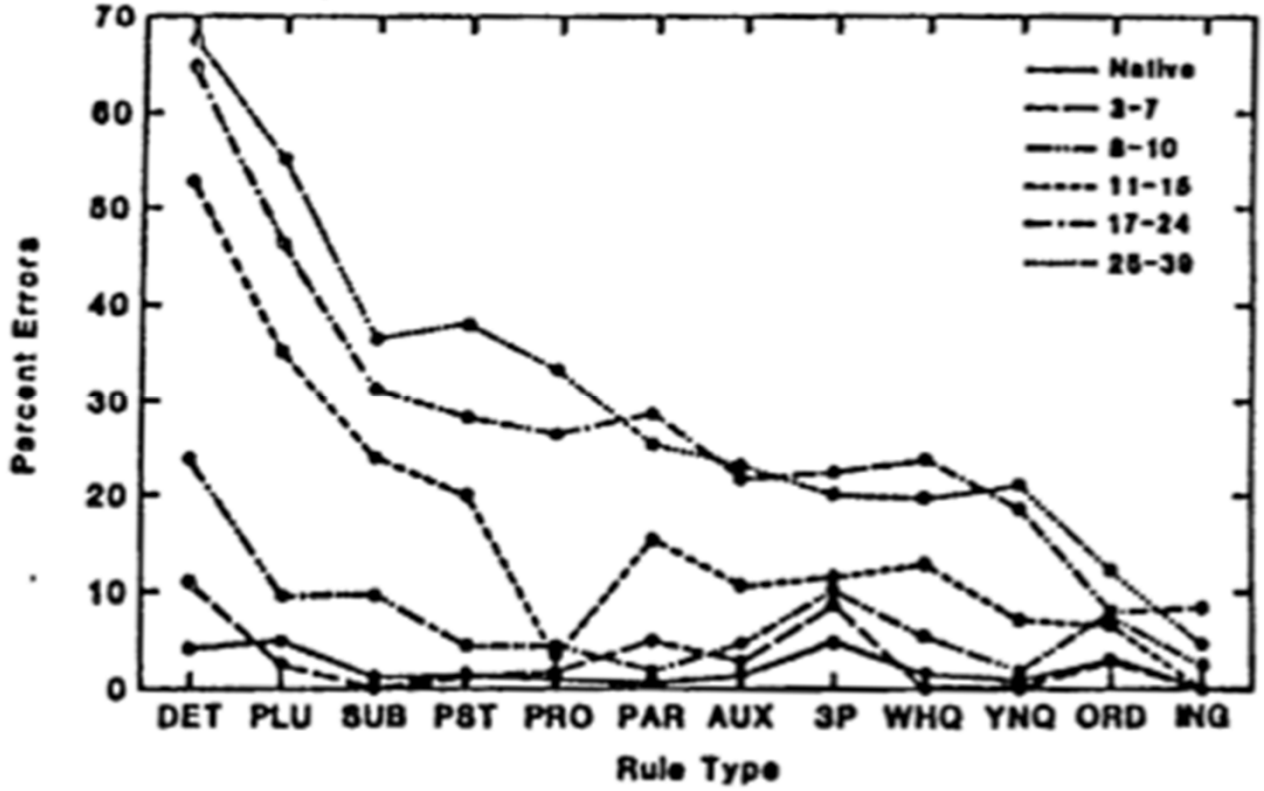


FIG. 3. Mean percentage of errors on 12 types of English rules.

Some of this looks like the order in which children learn language:

- progressive -ing and word order are easiest

Some could be transfer:

- Chinese has no determiners
- Kids learn questions very late, but they're among the easiest structures for late learners.

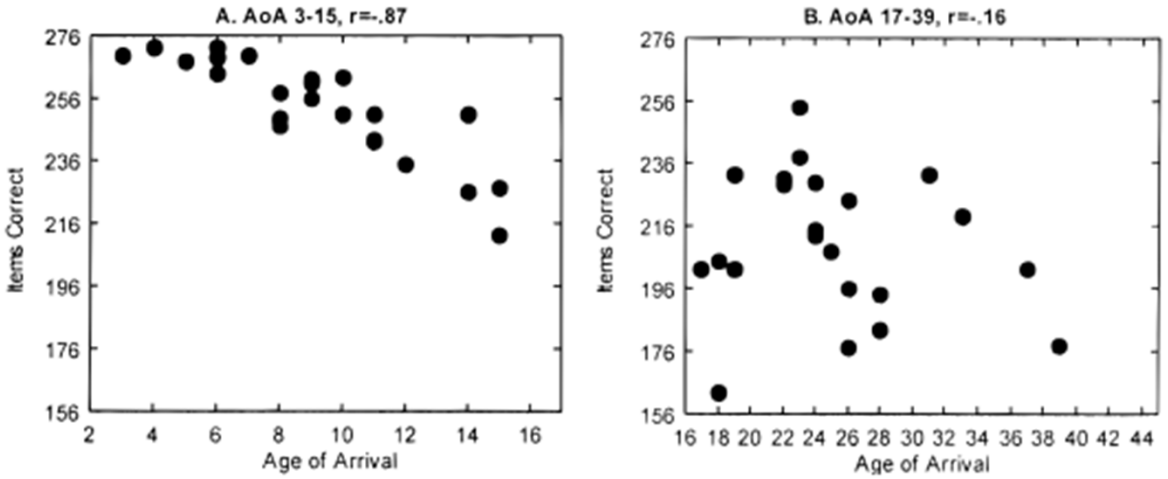


FIG. 1. The age function in J&N89, Early Arrivals (A), and Late Arrivals (B). In this and Figs. 2 and 3, points represent number of items correct for individual participants.

Chinese and Korean speakers learning English

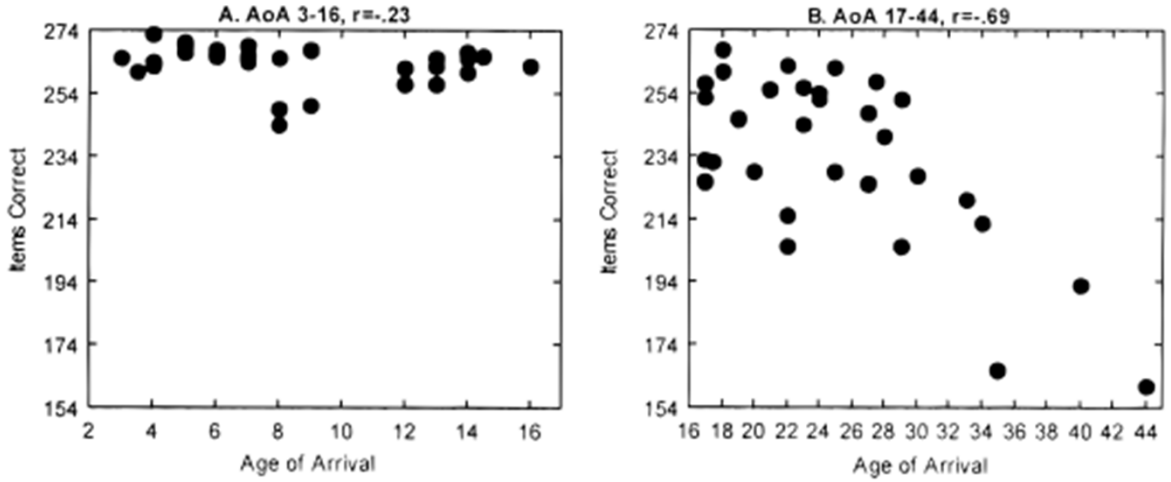


FIG. 2. The age function in the present study, Early Arrivals (A) and Late Arrivals (B).

Spanish speakers learning English

What are the consequences of this for the idea that AOA effects on GJT performance reflect *maturational constraints?*

Birdsong & Molis (2001, JML)

I’m not sure we can even call these “conclusions”

Overall, L2 attainment is lower for older learners.

This depends on:

- what constructions are tested
- how they are tested
- language pairings

It is hard to dissociate:

- age of acquisition from amount/quality of experience
- transfer effects from L2 acquisition
- performance from competence, generally speaking

For Tuesday

Read Menn, Chapter 8

(yes this is out of order!)